university of sussex • 21 october 2026

# Physics 2.0

## Formalisation and Physlib: a synergy between AI, mathematics and physics

**Joseph Tooby-Smith**
University of Bath

github.com/leanprover-community/physlib josephtoobysmith.com

# Advances in AI

**july 2026**

## The Jacobian conjecture

Keller, 1939. Counterexample in $\mathbb{C}^3$.
Alpoge, with Claude Fable 5

**august 2026**

## Ten open problems

From maths and CS
OpenAI “Astra”

**september 2026**

## Navier–Stokes problem

Finite time blow-up.
AI models

**october 2026**

## 372 problems

OpenAI, internal model

section 1 of 6

# Problems

**The problem**
Truth
Trust
Community
Recognition
Conclusion

# Some of the problems we face

1. AI is starting to do tasks humans used to do.
2. AI is starting to scoop human-led efforts at an unprecedented scale.
3. New inequalities are forming around AI access.
4. Current systems of recognition are not suited to this.

## The conclusion of this

# Physics 1.0 no longer works.

# What are we going to do about it?

*“Optimism is a duty. The future is open.”*

— Karl Popper

It is our duty as academics to be optimistic about the future, and to push academia to a place where it can best serve the needs of humanity in this new world.

## The goal of this talk

**To convince you that a solution for the future exists.**

## Things to preserve: four tenets of academia

**Truth**

**Trust**

**Community**

**Recognition**

# Physics 1.0

**truth**

Disseminated in academic papers.

**trust**

Peer review.

**community**

Spends its time writing papers, and reviewing them.

**recognition**

Citation counts, and numbers of papers.

section 2 of 6

# Reorganising truth

The problem

**Truth**

Trust

Community

Recognition

Conclusion

## Papers

# Do we want to keep academic papers?

## Papers

# Do we want to keep academic papers?

My answer: **No**.

# What is wrong with papers?

1. Hard to find information.
2. Hard to reuse.
3. Hard to find connections.
4. AI can write them, read them and review them.

# Pull towards structured data

**The Protein Data Bank**

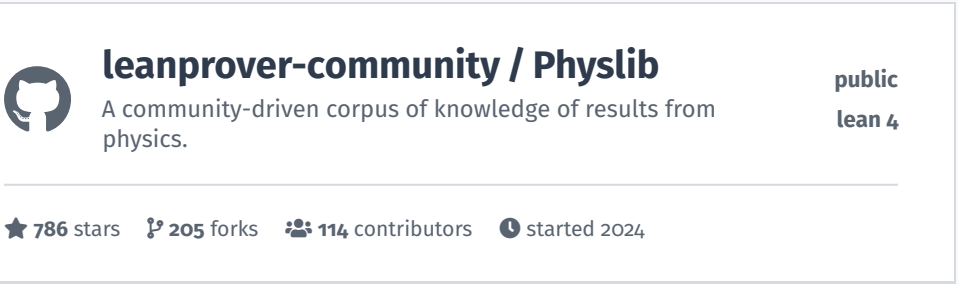
**AlphaFold**

Show structured data is beneficial for AI.

# Ingredient 1 of Physics 2.0

Replace papers with a **structured, open-source, community-run** corpus of knowledge.

# Good news, this exists…



# Dissent against our solution

1. Such a corpus of knowledge is hard to maintain.
2. It is hard to keep up to date.

AI helps with **all of this**.

section 3 of 6

# Adjusting trust

The problem
Truth
**Trust**
Community
Recognition
Conclusion

## Peer review

# Do we want to keep peer review?

## Peer review

# Do we want to keep peer review?

My answer: **Absolutely**. However…

# . . . there are problems with peer review

1. It does not scale efficiently with AI.
2. Mistakes get into papers, and slip past reviewers.

# A solution exists

Combine **peer review** with **automated checks on correctness** through **formalisation**.

# Technical interlude

# What is an interactive theorem prover?

A programming language in which you can write mathematical definitions, theorems, and their proofs, with mathematical **correctness** and **consistency** guaranteed.

# What is formalisation?

The process of writing results into an **ITP**.

# The four main interactive theorem provers

**Rocq** **Isabelle/HOL** **Lean** **Agda**

This talk is in **Lean**.

# A rapid overview of Lean

## Proof compiler

Definitions, theorems, proofs
⟶ **consistency & correctness, guaranteed**

## Built on type theory

Lean knows the **type** of everything, and checks that things have the type expected.

$$\underbrace{p}_{\text{proof}} : \underbrace{P}_{\text{theorem}} \qquad \underbrace{7}_{\text{term}} : \underbrace{\mathbb{N}}_{\text{type}}$$

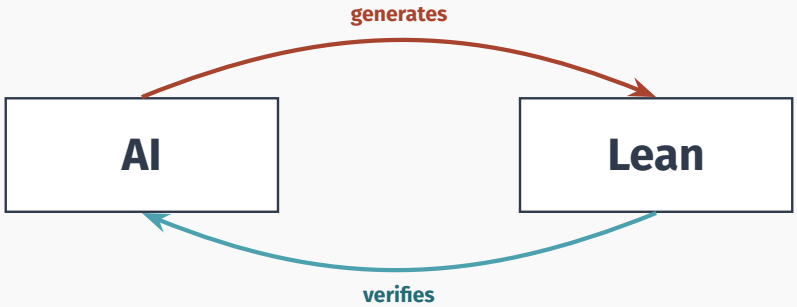
## An example

```
def energy (m v : Real) : Real :=
  m * v ^ 2 / 2

theorem energy_nonneg
    (m v : Real) (hm : 0 <= m) :
    0 <= energy m v := by
  unfold energy
  positivity
```

Open this example in the live editor

# The formalisation–AI loop



# End of technical interlude

# How good is AI-driven formalisation now?

It can essentially formalise **anything you want**.

Anthropic's Claude formalised Fermat's Last Theorem in Lean in about eleven days, on Lean's three standard axioms alone and with no `sorry`s.

# How good is AI-driven formalisation now?

It can essentially formalise **anything you want**.

Anthropic's Claude formalised Fermat's Last Theorem in Lean in about eleven days, on Lean's three standard axioms alone and with no `sorry`s.

However, this does not necessarily mean **well**.

# Formalisation does not give you complete trust

Formalisation is not the whole picture, and should not give you the whole trust. For that you also need peer review.

## Fermat's Last Theorem, made trivial

```
-- "the integers"
def MyInt : Type := Empty

def add (a b : MyInt) : MyInt :=
  Empty.elim a
def pow (a : MyInt) (n : Nat) : MyInt :=
  Empty.elim a

theorem flt (a b c : MyInt)
    (n : Nat) (hn : 3 <= n) :
    Ne (add (pow a n) (pow b n)) (pow c n) :=
  Empty.elim a
```

Lean accepts it. Nothing here is about numbers.

# Ingredient 2a of Physics 2.0

The **corpus of knowledge** for physics should have content which is **formally verified**, where possible.

**a dual process**

## Lean verification

the mathematics

**and**

## Human review

the physics context, style, organisation and quality

# Good news, this exists…



## leanprover-community / Physlib

A community-driven corpus of knowledge of results from physics, **formalised into Lean 4** where possible.

**public**

**lean 4**

**786** stars **205** forks **114** contributors started 2024

# . . . built for the whole physics community

**a choice of starting point**

There is no single correct axiomatisation, so we allow a **choice of starting point**.

# Physlib is one of three main ones



leanprover-community /

## mathlib4

The math library of Lean 4.

★ 4,309 ⑂ 1,768
👥 798 🕒 2021



leanprover-community /

## Physlib

Physics, digitalised into Lean. Community-driven.

★ 786 ⑂ 205
👥 114 🕒 2024



leanprover /

## CSLib

The computer science library of Lean.

★ 754 ⑂ 214
👥 50 🕒 2025

# Evidence that this can be done at scale and works

Top-level directories of `Physlib/`, by lines of Lean.

| | | | |
|---|---|---|---|
| Relativity | 27,545 | StatisticalMechanics | 2,449 |
| QFT | 20,162 | ProbabilisticTheory | 2,306 |
| Particles | 19,104 | Cosmology | 1,858 |
| SpaceAndTime | 16,069 | Meta | 1,773 |
| Mathematics | 15,795 | FluidDynamics | 1,361 |
| QuantumMechanics | 13,057 | CondensedMatter | 1,003 |
| ClassicalMechanics | 11,784 | ClassicalFieldTheory | 554 |
| Electromagnetism | 9,442 | Thermodynamics | 432 |
| StringTheory | 5,285 | Optics | 48 |
| Units | 4,076 | LatticeQFT | 22 |

**154,125** lines of Lean across **619** files.

## Evidence that this can be done at scale and works

# The Higgs boson, in Lean

live.lean-lang.org

# The problem

Lean verification takes the mathematics. But part of the trust still rests on **human review** — and that **cannot scale**.

# Ingredient 2b of Physics 2.0

Have part of the project which is similar to the arXiv, and has one-look review.

# Good news, this exists…

**PhyslibAlpha**

This is **PhyslibAlpha**, and it allows things to **scale**.

# Project structure

**upstream** **Mathlib**

**core**

## Physlib

Curated to a high standard for long-term reuse.

**rapid**

## PhyslibAlpha

Lighter review, for large-scale AI-assisted contributions.

**qi**

## QuantumInfo

Quantum information theory, being brought closer to Physlib.

**adjacent** **CSLib**

# Additional benefits of having Physlib

1. Allows verification of results, both human and AI.
2. Allows easy reuse, and improvement.
3. Allows assumptions to be tracked.
4. Allows one to easily explore and constrain model spaces.
5. Prevents bit-rot: one corpus, kept compiling.
6. Avoids duplicated work, and definitions that are incompatible with one another.
7. AI speed and efficiency makes this feasible.

section 4 of 6

# Community

The problem
Truth
Trust
**Community**
Recognition
Conclusion

## Community

# What will the community do now in this world?

# Community

## What will the community do now in this world?

My answer: They will **not** write and review papers.

# As mentioned earlier

Writing papers, and reviewing for journals, is exactly the work that AI is now **capable of**.

# Ingredient 3 of Physics 2.0

They will **build, maintain, review** and **grow** the corpus of knowledge (Physlib).

# Examples of activities

- Review
- Maintenance
- Refactoring
- Foundational work
- Formalisation
- Documentation
- Extending
- Adding experiments
- Building tools

# Overcoming difficulties in formalising physics

1. Validation does not come with the guarantees of verification.
2. One physical concept, many mathematical descriptions.
3. Physics papers aren't written as definitions and theorems.
4. How physical assumptions enter.

# Good news, people are already doing this.

**100+**
contributors

**7**
maintainers

**Always looking for more**

section 5 of 6

# Recognition

The problem
Truth
Trust
Community
**Recognition**
Conclusion

## Papers and citations

# Will people still get recognition based on papers and citations alone?

## Papers and citations

# Will people still get recognition based on papers and citations alone?

My answer: Of course **not**.

# The problem with current recognition

Recognition is currently based only on **publishing papers** and **citations**.

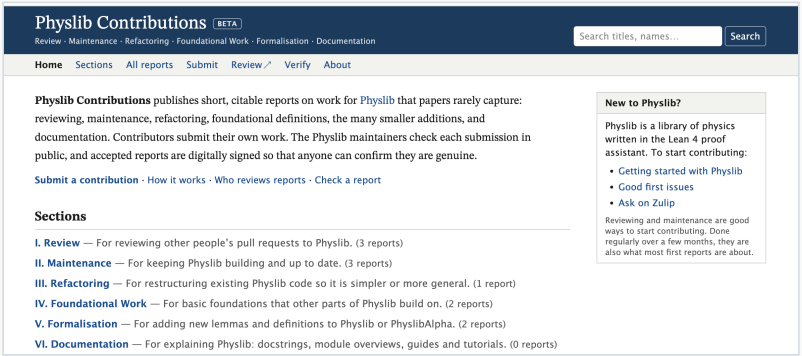
In this new world that is a **poor proxy** for the work that actually matters — because it is precisely the work we have seen AI is **very capable of**.

# Ingredient 4 of Physics 2.0

Recognition will be given to the work which **actually matters** in this new world.

# Good news, the infrastructure exists.

## contributions.physlib.io



section 6 of 6

# Conclusion

The problem
Truth
Trust
Community
Recognition
**Conclusion**

# Physics 2.0

**truth**

Papers → **corpus of knowledge**

**trust**

Peer review → **formalization + human review**

**community**

Writing papers → **Maintaining and growing the corpus**

**recognition**

Citations → **New age contributions**

# Call to action

**if you are hiring, or on grant committees**

Start to recognise contributions to projects like **Physlib**, **Mathlib** and **CSLib** on a par with papers.

**if you are an expert in physics**

Come to **Physlib**, help review, help push forward your area of expertise.

*"Optimism is a duty. The future is open. It is not predetermined. No one can predict it, except by chance. We all contribute to determining it by what we do. We are all equally responsible for its success."*

— Karl Popper, *The Lesson of This Century*

# Thank you for watching

**Joseph Tooby-Smith**
University of Bath

github.com/leanprover-community/physlib contributions.physlib.io josephtoobysmith.com